

2012
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Board-approved calculators may be used
- Write your Centre Number and Student Number at the top of pages 9, 13, 15, 17 and 19

Total marks – 100

Section I Pages 2–22

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–30
- Allow about 1 hour and 40 minutes for this part

Section II Pages 23–30

25 marks

- Attempt ONE question from Questions 31–35
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

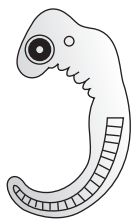
Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

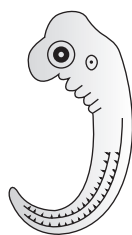
- 1 What is the name of the process that results in organisms containing DNA from different species?

(A) Transcription
(B) Transgenics
(C) Translation
(D) Translocation

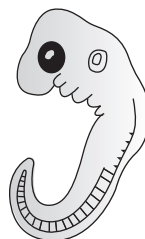
- 2 The diagram shows different vertebrate embryos at the same stage of development.



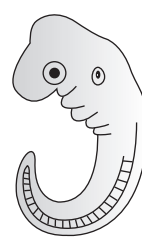
Fish



Amphibian



Bird

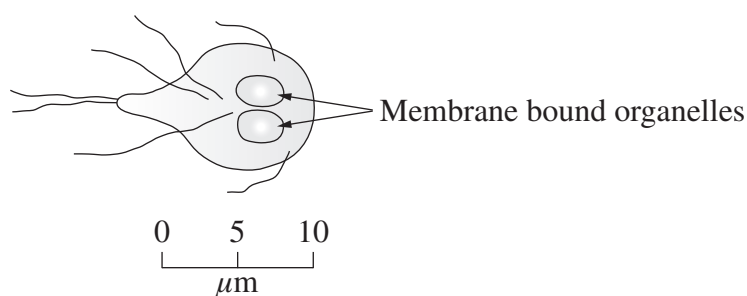


Human

How do the embryos provide evidence for evolution?

- (A) The embryos have different shaped eyes.
(B) Different adults evolve from the embryos.
(C) The embryos have structures that look similar.
(D) Divergent evolution results in common characteristics in the embryos.
- 3 Which of the following are all forms of defence that prevent the entry of pathogens into the body?
- (A) Cilia, sweat, saliva
(B) T cells, B cells, antibodies
(C) Inflammation, skin, phagocytosis
(D) Stomach acid, mucus, lymph system

- 4 The diagram shows a pathogen called Giardia.



What type of pathogen is Giardia?

- (A) Bacterium
- (B) Prion
- (C) Protozoan
- (D) Virus

5

Huntington's Disease is caused by an inherited gene that codes for a toxic protein.

Kwashiorkor is a disease caused by a deficiency of proteins in the body.

Mesothelioma is a disease caused by a gene mutation in the lungs after exposure to asbestos.

Which row in the table correctly classifies these diseases?

	<i>Huntington's Disease</i>	<i>Kwashiorkor</i>	<i>Mesothelioma</i>
(A)	Genetic	Nutritional	Environmental
(B)	Nutritional	Environmental	Environmental
(C)	Genetic	Nutritional	Genetic
(D)	Nutritional	Environmental	Genetic

- 6 How do vaccinations prevent disease?

- (A) They increase the inflammation process.
- (B) They enable the infected cells to seal off the pathogen.
- (C) They increase the number of antibodies against the pathogen.
- (D) They decrease the number of antigens that trigger the immune response.

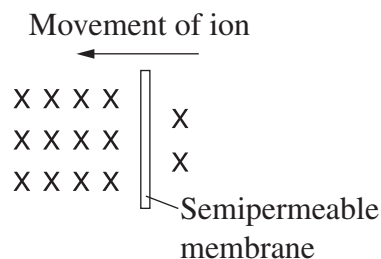
- 7 Why is it important to continue research into new antibiotics?
- (A) New prion diseases have been recently discovered.
 - (B) Resistant bacteria have evolved from the overuse of antibiotics.
 - (C) Viral infections require a broad range of antibiotics for eradication.
 - (D) New diseases are discovered regularly and all require new antibiotics.

- 8 Identical twins have the same genotype.

Why are there small differences between the phenotypes of identical twins?

- (A) Some genes are not co-dominant.
 - (B) Environment affects the expression of genes.
 - (C) Both parents are homozygous for those phenotypes.
 - (D) Chromosomes segregate independently during meiosis.
- 9 What feature can be used to distinguish mature xylem cells from mature phloem cells?
- (A) Phloem cells are located in vascular bundles.
 - (B) Phloem cells have a cytoplasm.
 - (C) Xylem cells are located in the leaves.
 - (D) Xylem cells have cell walls.

- 10 The diagram shows a model of the movement of ions (represented by X) across a semipermeable membrane.



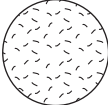
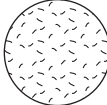
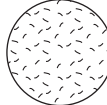
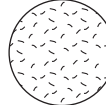
What type of process is modelled in the diagram?

- (A) Osmosis
- (B) Filtration
- (C) Diffusion
- (D) Active transport

- 11** Which of the following is an example of *hybridisation*?
- (A) The insertion of a bacterial gene for herbicide resistance into a cotton plant
 - (B) The culturing of a cell taken from the root of a carrot to form a small plant
 - (C) Artificial insemination of a domestic cat with wild cat semen to produce a Bengal cat
 - (D) A cutting taken from one variety of apple tree grafted onto the stem of a different variety of apple tree
- 12** Nitrogenous waste is at its highest concentration in
- (A) plasma in the renal vein.
 - (B) plasma in the renal artery.
 - (C) fluid in the collecting ducts of the kidney.
 - (D) interstitial fluid in the cortex of the kidney.
- 13** Why do organ transplants trigger an immune response in a recipient?
- (A) Antigens in the recipient's body recognise the organ as foreign.
 - (B) Cell surface markers on the organ attack the recipient's white blood cells.
 - (C) Antibodies in the recipient stimulate the production of antigens on the organ.
 - (D) The recipient's white blood cells recognise the antigens on the organ as foreign.
- 14** What is a role of the kidney in freshwater fish?
- (A) To remove water from the fish
 - (B) To absorb salt from the environment
 - (C) To excrete concentrated urine from the fish
 - (D) To decrease nitrogenous waste lost to the environment
- 15** Which of the following results in an increased absorption of water from the collecting tubule of the nephron?
- (A) An increase in the length of the collecting tubule
 - (B) A decrease in ADH released into the blood by the pituitary
 - (C) A decrease in glucose moving from the renal tubule into capillaries
 - (D) An increase in the concentration of dialysate solution in renal dialysis

- 16** A student carried out an investigation to identify the presence of microbes in water from different sources.

The student's lab notes are shown.

	<i>Control</i>	<i>Bottled water</i>	<i>Tap water</i>	<i>Tank water</i>
Inoculation of an agar plate with water sample	X	✓	✓	✓
Incubation at 37°C	✓	✓	✓	✓
Appearance of agar plate				

What can be inferred from these results?

- (A) The inoculation loop was not sterilised properly.
 - (B) The water from each of these sources is unsafe to drink.
 - (C) These water sources are contaminated with the same microbe.
 - (D) The agar plates were contaminated prior to the beginning of the experiment.
- 17** Which of the following correctly identifies the relationship between alleles, chromosomes and genes?
- (A) Genes contain chromosomes and alleles.
 - (B) Chromosomes contain genes but not alleles.
 - (C) Alleles are found in chromosomes but not in genes.
 - (D) Genes are parts of chromosomes and have different alleles.
- 18** How does a plant respond in order to keep cool on an extremely hot day?
- (A) It grows smaller leaves.
 - (B) It opens stomata in the leaves.
 - (C) It grows more hairs on the surface of the leaves.
 - (D) It decreases the number of stomata on the top of the leaves.

- 19** Haemoglobin provides an adaptive advantage to an endotherm in a cold environment because it allows
- (A) more oxygen to be dissolved in plasma.
 - (B) the organism to decrease its metabolic rate.
 - (C) more energy to be available to the organism.
 - (D) less carbon dioxide to be transported in the blood.
- 20** A student performed a first-hand investigation in an attempt to model natural selection.

Each week, a jar was refilled with a fresh packet of cream biscuits containing five different types for people to eat. After a month, there were mostly lemon cream biscuits in the jar.

What is the limitation of this investigation as a model for natural selection?

- (A) There is no variation in the 'species'.
- (B) Characteristics are not transmitted to successive generations.
- (C) Variants of the 'species' do not have the same chance of 'survival'.
- (D) Unfavourable characteristics are selected out of the population over time.

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2012 HIGHER SCHOOL CERTIFICATE EXAMINATION

Biology

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Centre Number

Section I (continued)

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Student Number

Part B – 55 marks

Attempt Questions 21–30

Allow about 1 hour and 40 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (5 marks)

You performed a first-hand investigation to estimate the size of blood cells.

- (a) How did you estimate the size of the cells?

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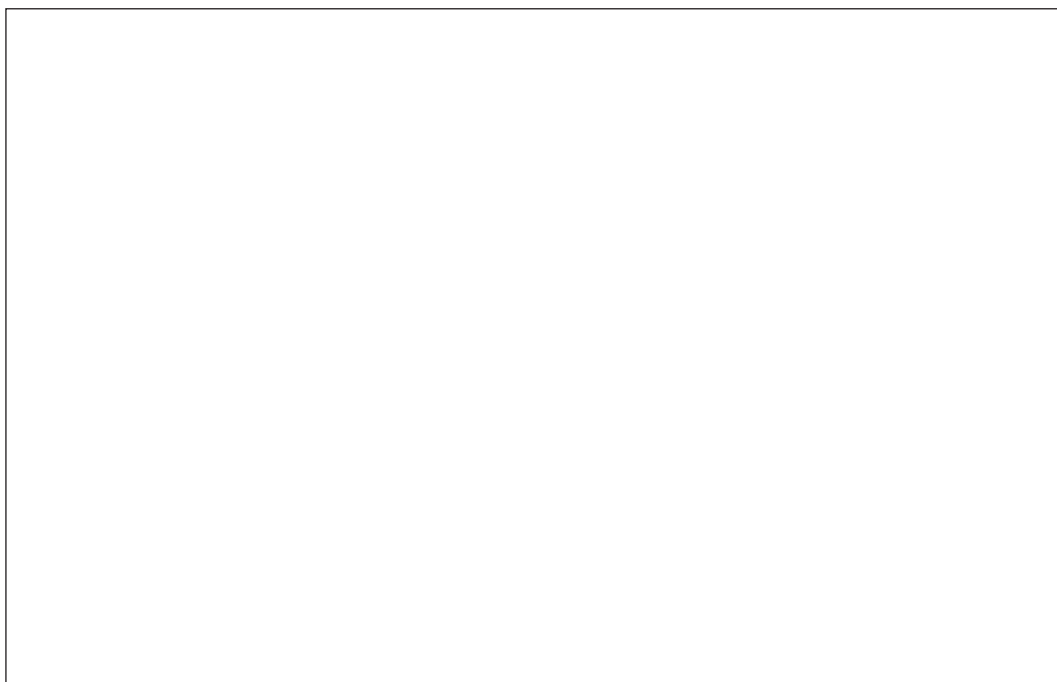
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- (b) Draw a scaled diagram that shows the features of both a red blood cell and a white blood cell.

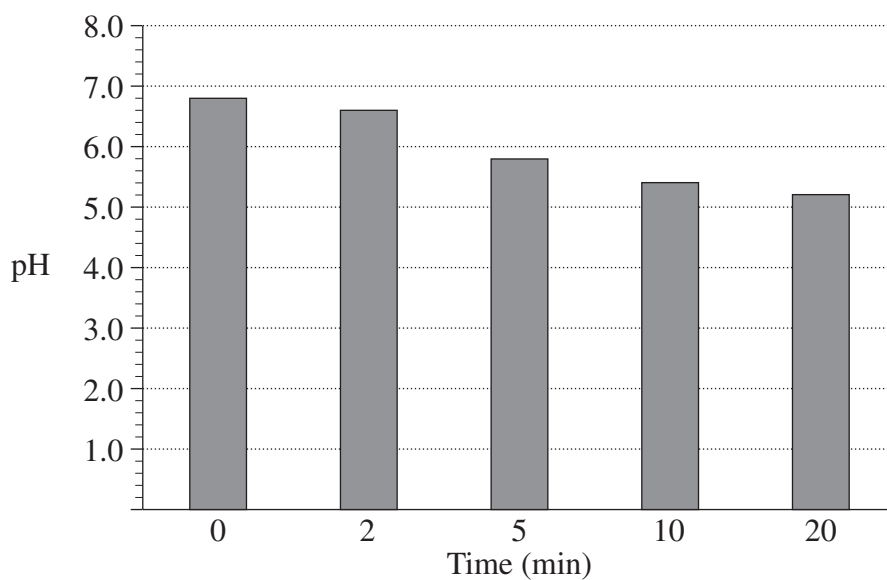
3



Question 22 (7 marks)

Students in a class conducted a first-hand investigation to test the hypothesis that if CO_2 were continually bubbled in water then the pH would decrease over time.

One student presented the data in the graph, as shown.



(a) Present these data in a table.

2

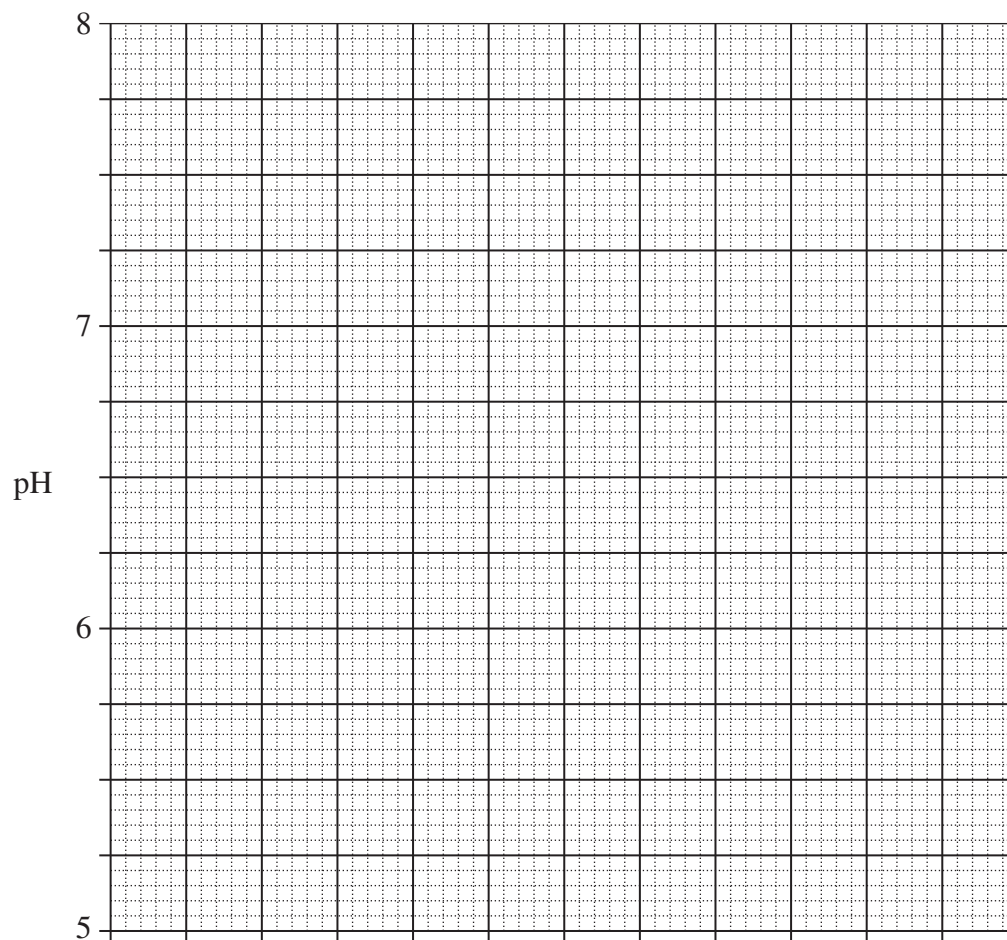
Time (min)	pH
0	6.8
2	6.6
5	5.8
10	5.4
20	5.2

Question 22 continues on page 11

Question 22 (continued)

- (b) On the grid below, plot the data from the table in part (a) and draw a curve of best fit.

3



- (c) Why is it better to represent these results as a curve of best fit rather than as a column graph?

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End of Question 22

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Biology

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Centre Number

Section I – Part B (continued)

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Student Number

Question 23 (5 marks)

A non-infectious disease was observed in a mother and her four sons who live with her. She has no daughters. The father of these children does not have the disease and does not live with them. The woman's parents and her two sisters who live overseas do not have the disease.

- (a) Her doctor suspects that the disease is NOT inherited. Identify data that could be collected to investigate a non-inheritable cause of the disease. 1

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- (b) A geneticist suspects that the disease is inherited.

- (i) Draw the family pedigree for this disease. 2

- (ii) From the evidence, what indicates that the disease could be the result of a recessive allele and not be sex linked?

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Question 24 (5 marks)

You conducted first-hand investigations to test the effects of temperature, pH and substrate concentration on enzyme activity.

- (a) Complete the following table by identifying the variables for ONE of your investigations. **1**

<i>Independent variable</i>	<i>Dependent variable</i>	<i>Kept constant</i>
.....		pH, substrate concentration, enzyme concentration

- (b) Outline how you measured the activity of an enzyme in your investigation. In your answer, name the enzyme. **2**

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- (c) Describe how a condition needed for optimal enzyme activity would be expected to vary between endotherms and ectotherms. **2**

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Centre Number

Section I – Part B (continued)

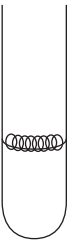
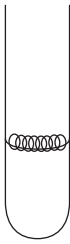
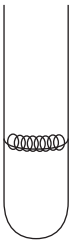
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Student Number

Question 25 (6 marks)

Beadle and Tatum's experiment involved the analysis of bread mould growth. Bread mould uses an enzyme to make the amino acid arginine.

The diagram shows bread mould growth after culturing on two different media. In one part of the experiment, the bread mould had been irradiated before culturing.

Key	Without irradiation		With irradiation	
		Bread mould growth		
				
	Minimal medium	Minimal medium + arginine	Minimal medium	Minimal medium + arginine

- (a) How do these results support the 'one gene – one protein' hypothesis?

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- (b) Justify the types of secondary sources that would be acceptable to use to research Beadle and Tatum's experiment.

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Question 26 (5 marks)

A scientist performed an epidemiological study to investigate the cause and effect relationship of smoking and lung cancer as follows:

5

1. Handed out a scientifically valid questionnaire to all colleagues (n = 144) at work
2. Checked that there were an equal number of male and female respondents
3. Discovered that there were more non-smoking respondents than smoking respondents. Removed some of the non-smokers until both groups had equal numbers
4. Checked that all the respondents had a medical check-up in the past year
5. Analysed data, wrote the paper and published it in a scientific blog.

From the information provided, analyse the methodology used by this scientist.

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2012 HIGHER SCHOOL CERTIFICATE EXAMINATION

Biology

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Centre Number

Section I – Part B (continued)

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Student Number

Question 27 (4 marks)

During a major international horse event in Australia, a group of horses, including some from overseas, is discovered to be infected by a deadly virus. This virus is only found in Australia.

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Give reasons for strategies that could be carried out to control this disease outbreak.

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Question 28 (5 marks)

Explain the relationship between replication of DNA and evolution.

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2012 HIGHER SCHOOL CERTIFICATE EXAMINATION

Biology

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Centre Number

Section I – Part B (continued)

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Student Number

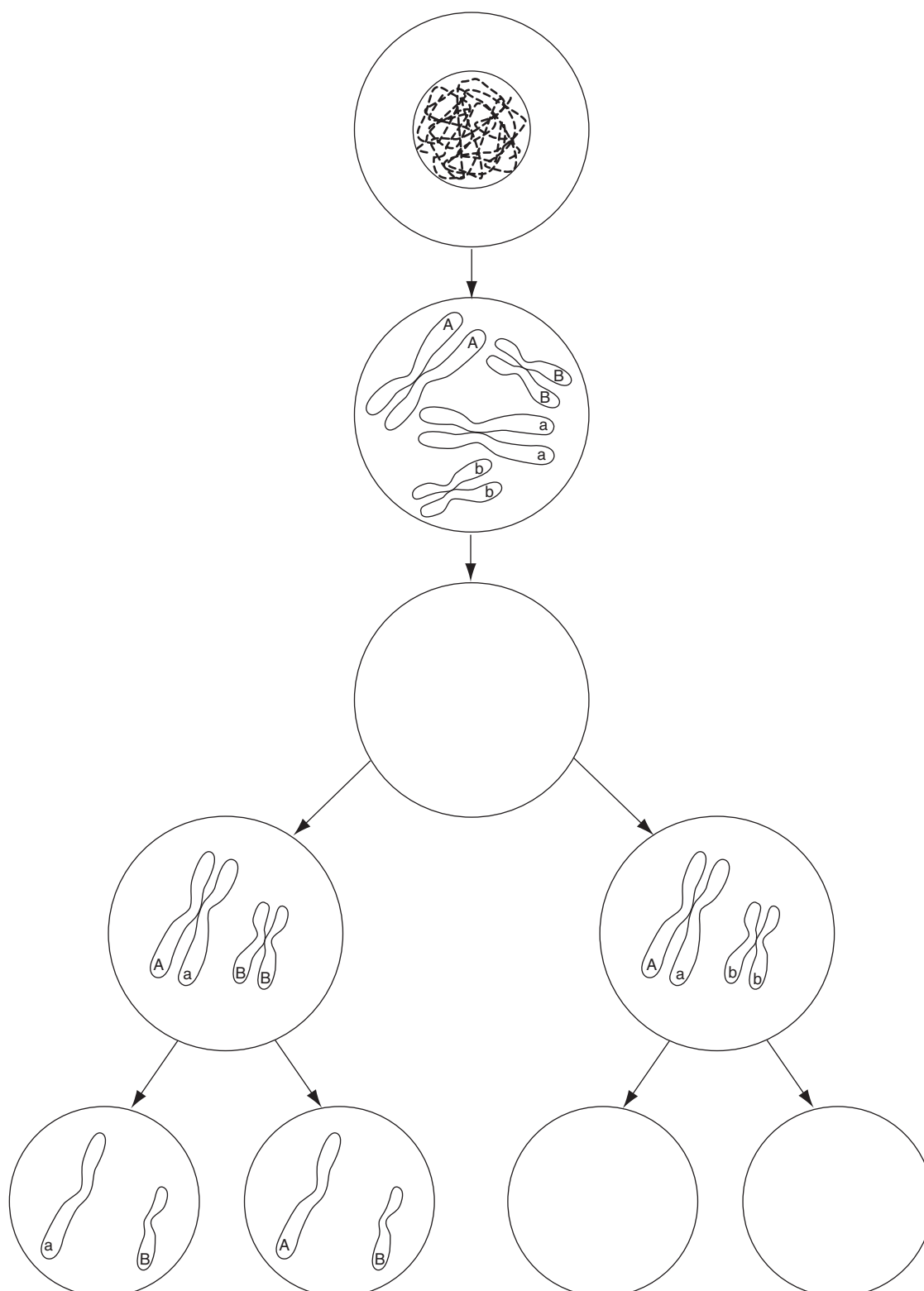
Question 29 (5 marks)

Please turn over

Question 29 (5 marks)

- (a) Complete the following diagram to show the process by which gametes are formed.

3



Question 29 continues on page 21



Question 29 (continued)

- (b) How does the segregation of chromosomes during meiosis lead to a wide variety of gametes being produced? **2**

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End of Question 29

Please turn over



8

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Biology

Section II

25 marks

Attempt ONE question from Questions 31–35

Allow about 45 minutes for this section

There are four Section II Answer Booklets labelled Part (a), Part (b), Part (c) and Part (d).

Answer each part of the question in the relevant Answer Booklet.

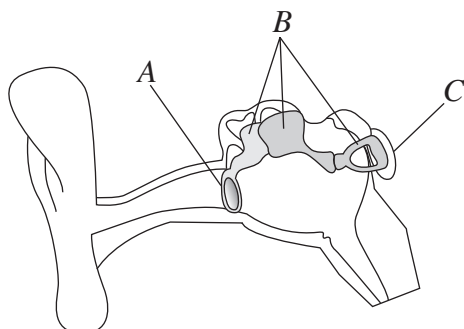
Extra writing booklets are available.

	Page
Question 31 Communication	24–25
Question 32 Biotechnology	26
Question 33 Genetics: The Code Broken?	27
Question 34 The Human Story	28–29
Question 35 Biochemistry	30

Question 31 — Communication (25 marks)

Answer part (a) in Section II Answer Booklet – Part (a).

(a) Parts of the ear are shown below.



(i) Identify the parts *A*, *B* and *C*.

2

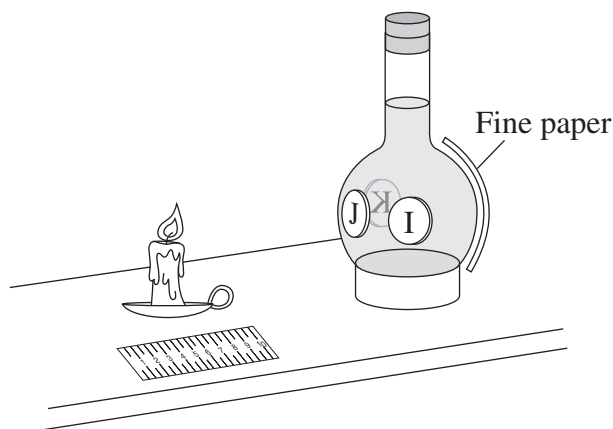
(ii) Outline the functions of part *B*.

2

Answer part (b) in Section II Answer Booklet – Part (b).

(b) An experiment was performed to model accommodation in the eye:

- a round bottom flask was filled with a solution
- the flask was placed on a stand on which it was able to rotate
- three lenses (I, J and K) of different thicknesses were attached to the surface of the flask
- a layer of fine paper was stuck around the opposite side of the flask
- a candle was placed near the flask
- different lenses could be brought into the light path by rotating the flask
- an image of the candle was observed on the layer of fine paper.



Question 31 continues on page 25

Question 31 (continued)

- (i) Match THREE pieces of apparatus used in this experiment with THREE identified parts of the eye. **2**
- (ii) Describe the quantitative data that could be collected in the experiment. **2**
- (iii) Compare this model to mechanisms of accommodation in the eye. **2**

Answer part (c) in Section II Answer Booklet – Part (c).

- (c) (i) Relate specialised features of the eyes of TWO named animals to their environment. **2**
- (ii) Describe the detection of vibrations by fish and insects, and the events which lead to a response. **5**

Answer part (d) in Section II Answer Booklet – Part (d).

- (d) Demonstrate how technologies help correct specific biological problems associated with human communication. **8**

End of Question 31

Question 32 — Biotechnology (25 marks)

Answer part (a) in Section II Answer Booklet – Part (a).

- (a) With reference to TWO biotechnological practices that use yeast, copy and complete the following table in your answer booklet. **4**

<i>Biotechnological practice</i>	<i>Name of chemical produced by yeast</i>	<i>Purpose of chemical produced</i>

Answer part (b) in Section II Answer Booklet – Part (b).

- (b) (i) Explain the purpose of a named transgenic organism. **3**
- (ii) Construct a diagram that demonstrates the sequence of events that results in the formation of recombinant DNA in a transgenic organism. **3**

Answer part (c) in Section II Answer Booklet – Part (c).

- (c) (i) Why could aquaculture be considered a biotechnology? **2**
- (ii) Compare aquaculture with early biotechnologies applied to agriculture. **5**

Answer part (d) in Section II Answer Booklet – Part (d).

- (d) Explain how a detailed understanding of cell chemistry has resulted in society making more effective use of biotechnology. Support your answer with examples. **8**

Question 33 — Genetics: The Code Broken? (25 marks)

Answer part (a) in Section II Answer Booklet – Part (a).

- (a) With reference to TWO types of cloning, copy and complete the following table in your answer booklet. **4**

<i>Type of cloning</i>	<i>Process used</i>	<i>Example</i>

Answer part (b) in Section II Answer Booklet – Part (b).

- (b) (i) Construct a model of a section of a double stranded DNA molecule containing the sequence GAT using a selection of the symbols below. Include a key in your answer. **3**

Ⓒ ⒫ Ⓔ Ⓐ Ⓖ Ⓓ ⓤ

- (ii) Outline how genes are expressed. **3**

Answer part (c) in Section II Answer Booklet – Part (c).

- (c) (i) What is the Human Genome Project? **2**
- (ii) Explain how recombinant DNA technologies can be used to identify the position of a gene on a chromosome. **5**

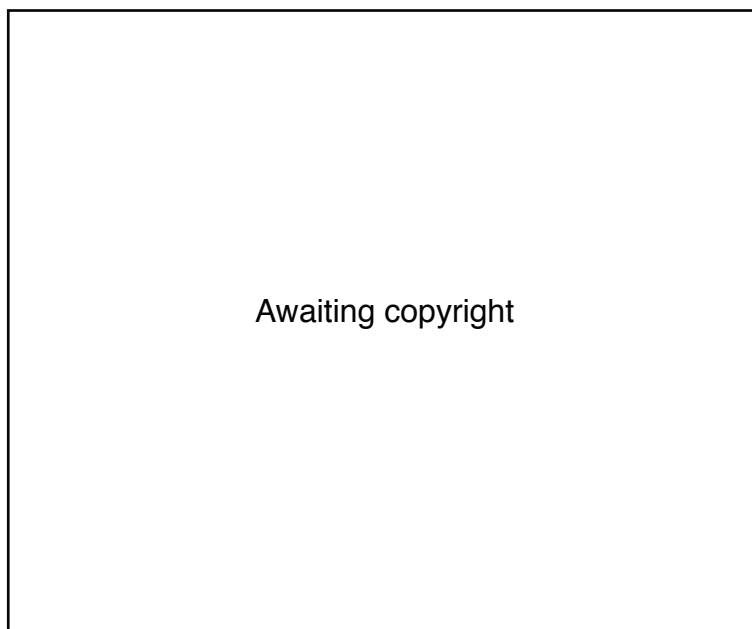
Answer part (d) in Section II Answer Booklet – Part (d).

- (d) Compare and contrast the effects of germ line mutation and transposable genetic elements on whole organisms. **8**

Question 34 — The Human Story (25 marks)

Answer part (a) in Section II Answer Booklet – Part (a).

- (a) (i) Using the diagram, name TWO features of the *Homo sapiens* skeleton which make it unique among the primates. **2**



- (ii) Outline how ONE named non-skeletal characteristic has assisted primates in their evolution. **2**

Question 34 continues on page 29

Question 34 (continued)

Answer part (b) in Section II Answer Booklet – Part (b).

- (b) Examine the data collected on *Homo neanderthalensis* fossils found in recent years.

<i>Date</i>	<i>Data</i>
2008	• Full sequence of mitochondrial DNA of a range of Neanderthal fossils from within and between different fossil sites • The sequence is nearly identical within one fossil site • The sequence is very different between fossil sites
2010	• Full sequence of Neanderthal nuclear DNA • 1%–4% of genes in European modern humans are specific Neanderthal genes • No identifiable specific Neanderthal genes in modern sub-Saharan African humans • No specific modern human genes in nuclear DNA of Neanderthal fossils

- (i) What inferences can be made about Neanderthal populations, based on the data collected in 2008? **3**
- (ii) What inferences can be made about migration and breeding, based on the data collected in 2010? **3**

Answer part (c) in Section II Answer Booklet – Part (c).

- (c) How have polymorphism and clinal gradation contributed to the diversity of human populations? In your answer, use at least one human phenotype as an example. **7**

Answer part (d) in Section II Answer Booklet – Part (d).

- (d) (i) For TWO named hominids, describe fossil evidence that infers the use of technology which led to cultural development. **2**
- (ii) Predict and justify future directions of human biological evolution in the context of TWO technologies already developed by modern humans. **6**

End of Question 34

Question 35 — Biochemistry (25 marks)

Answer part (a) in Section II Answer Booklet – Part (a).

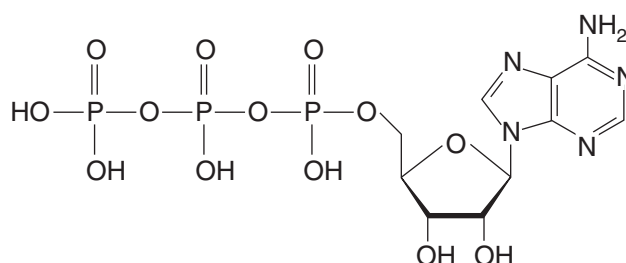
- (a) (i) Name a product of photosynthesis and its chemical source. **2**
- (ii) Outline how photosynthesis could be used to replace materials presently obtained from non-renewable resources. **2**

Answer part (b) in Section II Answer Booklet – Part (b).

- (b) You performed a first-hand investigation to determine the effect of light intensity on production of gas in a suitable pond weed.
- (i) Draw a labelled diagram of the experimental set-up. **2**
- (ii) Describe the quantitative data you collected in the experiment. **2**
- (iii) How would you alter this investigation to measure the effect of carbon dioxide on photosynthesis? **2**

Answer part (c) in Section II Answer Booklet – Part (c).

- (c) A diagram of adenosine triphosphate (ATP) is shown.



- (i) With reference to the diagram, describe how ATP performs its biological function. **2**
- (ii) Explain how light energy is converted to ATP and NADPH during the light-dependent reaction. **5**

Answer part (d) in Section II Answer Booklet – Part (d).

- (d) Explain how knowledge of biochemical pathways has been enhanced by an understanding of radioactive isotopes. **8**

End of paper



2012 HSC Biology Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	A
4	C
5	A
6	C
7	B
8	B
9	B
10	D
11	C
12	C
13	D
14	A
15	A
16	D
17	D
18	B
19	C
20	B

Section I, Part B

Question 21 (a)

Criteria	Marks
- Identifies that cells are viewed by a microscope or equivalent AND - Correctly states a mechanism or piece of equipment for measuring size of cells	2
One of the above	1

Question 21 (b)

Criteria	Marks
Correctly draws the cells at the right relative scales and appropriate detail and has a correct scale as part of the drawing	3
- Draws the cells identifiable as the appropriate types OR - Draws one of the cells that is identifiable and includes a correct scale	2
- Draws a cell of an identifiable type OR - Gives correct information about the size of blood cells	1

Question 22 (a)

Criteria	Marks
Constructs a table that includes ALL of the following: headings, units and ordered data	2
Constructs a table that includes TWO of the following: headings, units and ordered data	1

Question 22 (b)

Criteria	Marks
Includes ALL of the following: X-axis correctly labelled; scaled (linear); points plotted correctly from data table; curve of best fit correct	3
Includes any THREE of the above	2
Includes any ONE of the above	1

Question 22 (c)

Criteria	Marks
• States that the independent variable is continuous, and not linear	2
• States ONE of the above	1

Question 23 (a)

Criteria	Marks
• Provides any relevant data	1

Question 23 (b) (i)

Criteria	Marks
• Constructs an appropriate pedigree using accepted symbols	2
• Constructs an appropriate pedigree without accepted symbols OR • Constructs a pedigree with some correct information	1

Question 23 (b) (ii)

Criteria	Marks
• Provides ONE piece of evidence that indicates why the gene is recessive AND • Provides ONE piece of evidence that indicates why the gene cannot be sex linked	2
• Provides ONE of the above or does not link the correct answer with the condition (recessive or sex linked)	1

Question 24 (a)

Criteria	Marks
• Correctly identifies the independent and dependent variables	1

Question 24 (b)

Criteria	Marks
Correctly identifies a named enzyme and describes how its activity is measured	2
- Names an enzyme OR - Explains in general terms how the activity of an enzyme might be measured	1

Question 24 (c)

Criteria	Marks
Demonstrates a difference in enzyme activity between endotherms and ectotherms	2
Identifies one difference between ectotherms and endotherms	1

Question 25 (a)

Criteria	Marks
- Identifies a link between X-rays and mutation - Identifies the result and links it to a gene mutation - Identifies the gene mutation as being related to arginine synthesis - Correctly links these ideas to the ‘one gene–one protein’ hypothesis	4
Any THREE of the FOUR above	3
Any TWO of the FOUR above	2
- Any ONE of the FOUR above OR - Provides some information about the ‘one gene–one protein’ hypothesis.	1

Question 25 (b)

Criteria	Marks
Gives reasons for the appropriateness of TWO or more secondary sources	2
- Names TWO appropriate secondary sources OR - Names ONE source with relevant reason	1

Question 26

Criteria	Marks
- Analyses the methodology of epidemiology - Identifies strengths and/or weaknesses in each part of the study from the information provided	5
- Analyses the methodology of epidemiology - Identifies strengths and/or weaknesses in most but not all parts of the study (does not address all points of the study) from the information provided	4
- Demonstrates a basic knowledge of the methodology of epidemiology - Identifies strengths or weaknesses in some parts of the study OR identifies strengths and weaknesses in one part of the study	3
- Recalls a feature of epidemiology - Identifies a weakness or a strength of the method	2
- Identifies a feature of epidemiology OR - Identifies a weakness or a strength of the method	1

Question 27

Criteria	Marks
Supports at least TWO strategies with an argument for each	4
Identifies TWO appropriate strategies with one supporting reason	3
- Identifies TWO appropriate strategies with no supporting reasons OR - Supports at least ONE strategy with an argument	2
Identifies ONE appropriate strategy with no supporting reason	1

Question 28

Criteria	Marks
• Demonstrates an understanding of the biological concepts that link DNA replication to sexual and asexual reproduction • Links changes to DNA during DNA replication to the generation of variants in a species • Relates new variants which arise to changes in gene frequencies of a population as a result of natural selection • Identifies the above as evolution	4–5
• Demonstrates an understanding of the biological concepts that link DNA replication to sexual or asexual reproduction • Links changes to DNA during DNA replication to the generation of variants in a species • Identifies that variation in the population will result in changes in the species due to natural selection	3
• Demonstrates an understanding of the biological concepts that link DNA replication to sexual or asexual reproduction AND • Links changes to DNA during DNA replication to the generation of variants in a species OR identifies that variation in the population will result in changes in the species due to natural selection	2
• Provides some relevant information about DNA replication OR • Provides some relevant information about evolution	1

Question 29 (a)

Criteria	Marks
• Diagram 1: tetrad line up correctly, alleles in correct position, homologous chromosomes paired and joined correctly and crossing over has occurred • Diagram 2: haploid, with appropriate chromosomes and alleles in both	3
• Diagram 1 as above OR • TWO elements of Diagram 1 and Diagram 2 as above	2
• Provides correct information about ONE element in correct cell	1

Question 29 (b)

Criteria	Marks
• Demonstrates a sound knowledge of independent assortment during gamete formation and relates to a range of gamete combinations	2
• Provides some information about independent assortment	1

Question 30

Criteria	Marks
- Communicates succinctly, logically and sequentially using precise biological terms - Provides a detailed knowledge and understanding of biological methodology and concepts and is able to adequately link these to the nature and prevention of disease - Identifies biological research relationships between the named scientists - Provides a clear understanding of the historical development of biological concepts and their applications and implications for society and the environment. - Provides an understanding of valid experimental processes or appropriate technologies for developing biological ideas and their application to society	7–8
- Communicates effectively using biological terms - Provides some knowledge and understanding of biological methodology and concepts and is able to adequately link some of these to the nature and prevention of disease - Provides an understanding of some historical developments of biological concepts and their applications and implications for society and the environment AND - Identifies biological research relationships between the named scientists OR links experimental processes and appropriate technologies to their application to society as related to the topic.	5–6
- Communicates using clear written expression - Provides knowledge and understanding of some biological methodology and concepts and is able to show some linkage between them - Provides some understanding of historical developments of biological concepts AND - Identifies biological relationships between the named scientists OR identifies a named scientist and shows how their findings have assisted our understanding of the nature and prevention of disease OR links experimental processes and appropriate technologies to their application to society as related to the topic	3–4
Provides a limited understanding and knowledge of the topic areas	2
Provides some relevant information	1

Question 31 (a) (i)

Criteria	Marks
• All THREE structures correctly named	2
• Names at least ONE structure correctly	1

Question 31 (a) (ii)

Criteria	Marks
• Clearly indicates the functions of the identified structure	2
• Provides some information about the function of the identified structure	1

Question 31 (b) (i)

Criteria	Marks
• Corresponds THREE ocular structures with the experimental apparatus	2
• Corresponds at least ONE ocular structure with the experimental apparatus	1

Question 31 (b) (ii)

Criteria	Marks
• Provides the characteristics of the ‘quantitative data’	2
• Provides some information about the ‘quantitative data’	1

Question 31 (b) (iii)

Criteria	Marks
• Shows similarities and/or differences between the model and the eye	2
• Shows a similarity or a difference between the model and the eye	1

Question 31 (c) (i)

Criteria	Marks
Outlines features of the eyes of TWO animals and relates features to their environment	2
Outlines features of the eyes for TWO animals OR Outlines features of the eyes for ONE animal or relates features to their environment	1

Question 31 (c) (ii)

Criteria	Marks
- For both fish and insects identifies the structures for detecting vibration and includes their location, nature and gives some examples - Outlines the process from detection of vibration to response - Provides some information about frequencies and identification of the source of the sound	4-5
- Provides some information about structures for detecting vibration in insects and fish - Outlines the process from detection of vibration to response OR provides some information about frequencies and identification of the source of the sound	3
- Provides some information about structures for detecting vibration in insects or fish - Outlines the process from detection of vibration to response or provides some information about frequencies and identification of the source of the sound OR - Provides some information about structures for detecting vibration in insects and fish OR - Outlines the process from detection of vibration to response and provides some information about frequencies and identification of the source of the sound	2
Provides some relevant information about detection OR processing of vibrations in insects or fish	1

Question 31 (d)

Criteria	Marks
- Shows in detail a range of biological problems associated with communication and links these to technological measures that have been used to correct them - Identifies different strategies for correcting the problems - Articulates ideas in a coherent and logical manner and includes the correct use of scientific principles and ideas	7–8
- Shows some biological problems associated with communication and links these to technological measures that have been used to correct them - Identifies different strategies for correcting some of these - Articulates ideas in a coherent and logical manner and includes the correct use of scientific principles and ideas	5–6
- Shows a biological problem associated with each of two modalities of communication and identifies a technological measure to correct each of them OR - Shows a biological problem associated with a modality of communication and identifies technological measures and a strategy to correct it	3–4
- Shows a biological problem associated with communication and outlines a technological measure to correct it OR - Identifies technological measures for correcting communication problems but does not link them to the biological problem	2
Provides some information relevant to human communication or its treatment	1

Question 32 (a)

Criteria	Marks
Identifies two biotechnological practices that use yeast, the chemicals produced and their purpose	4
Identifies two biotechnological practices that use yeast, includes product and purpose for one and product or purpose for the other	3
- Identifies the product and purpose of one named biotechnological practice that uses yeast OR - Identifies two different biotechnological practices that use yeast	2
Identifies some information about a biotechnological practice, product or purpose for the use of yeast	1

Question 32 (b) (i)

Criteria	Marks
• Clearly provides the relationship between a transgenic organism and its purpose	3
• Names a transgenic organism and its purpose	2
• Provides some information about a transgenic organism	1

Question 32 (b) (ii)

Criteria	Marks
• Diagram shows a detailed description of the major steps required to produce a named transgenic species	3
• Conveys an adequate description of the major steps required to produce a named transgenic species	2
• Provides information about producing a transgenic species	1

Question 32 (c) (i)

Criteria	Marks
• Relates manipulation of the environment and/or the species to provide a product for humans	2
• Defines biotechnology OR • Provides some information about aquaculture	1

Question 32 (c) (ii)

Criteria	Marks
- For aquaculture and traditional agriculture, identifies key features for both that are common and/or different - Provides specific examples that highlight the biotechnologies being used.	4-5
- Provides some information about features that are common and/or different in both aquaculture and traditional agriculture - Gives an example of a biotechnology being used	3
- Provides some information about aquaculture or traditional agriculture and gives an example of a biotechnology being used OR - Provides some information about aquaculture and traditional agriculture	2
Provides some relevant information about aquaculture or traditional agriculture	1

Question 32 (d)

Criteria	Marks
- Clearly identifies components of early biotechnology and relates how knowledge of cell biochemistry has implications for specific targeting of biotechnology in the modern context - Relates an extensive and detailed understanding of cellular biochemical processes to the development of several named biotechnological processes - Relates the implications of specific biochemical processes to the development of specific biotechnological products - Articulates ideas in a coherent and logical manner and includes the correct use of scientific principles and ideas	7–8
- Identifies at least ONE component of early biotechnology and shows how knowledge of cell biochemistry has changed biotechnology - Relates in general terms the understanding of cellular biochemistry to the development of modern biotechnology processes OR - Provides an extensive understanding of cellular biochemical processes and relates them to several named biotechnological processes	5–6
- Identifies a component of early biotechnology - Relates a biochemical process to a modern biotechnology OR - Relates in general terms an understanding of cellular biochemical processes to modern biotechnology processes	3–4
Provides some relevant information about cell biochemistry and a biotechnology	2
Provides some relevant information that identifies a link between cell biochemistry or biotechnology	1

Question 33 (a)

Criteria	Marks
Identifies processes used and specific examples for TWO types of cloning	4
Identifies two named cloning processes, includes process used and specific example for one type of cloning and process or specific example for the second type	3
- Identifies a cloning process and example of ONE named type of cloning OR - Partially describes two named types of cloning	2
Identifies TWO of: a type of cloning, purpose or example	1

Question 33 (b) (i)

Criteria	Marks
- Provides the main features of DNA structure - Provides a key for the model	3
- Provides some features of DNA structure - Provides a key for the model	2
- Provides some information about DNA structure OR - Provides a key	1

Question 33 (b) (ii)

Criteria	Marks
Indicates main features of the process and communicates it succinctly and logically	3
Indicates main features of the process	2
Provides some information about the process	1

Question 33 (c) (i)

Criteria	Marks
Lists at least two main features of the HGP	2
Lists one feature of the HGP	1

Question 33 (c) (ii)

Criteria	Marks
- Provides a good understanding of recombinant DNA technologies - Identifies the need for a probe - Identifies the labelling of the probe as a necessity - Identifies that the sequence of the gene of interest must exist on an intact chromosome	4–5
- Identifies the need for a probe AND the necessity of a labelling method OR - Provides a list of techniques that can be used but does not link	2–3
Provides some information on recombinant DNA technologies that can be used	1

Question 33 (d)

Criteria	Marks
- Provides an in-depth understanding of germ line mutation and reasons for its impact on whole organism phenotypes - transmitted to offspring - all cells of offspring contain mutation - Provides an in-depth understanding of transposons in their disruption of genes in genomes - insertion destroys gene - replication and relocation increases effect - chromosome elongation can disrupt cell division - Articulates in a coherent and logical manner and includes the correct use of scientific principles and ideas	7–8
- Provides some understanding of germ line mutation and reasons for its impact on whole organisms - Provides some understanding of transposons in their disruption of genes in genomes - Articulates in a coherent and logical manner and includes the correct use of scientific principles and ideas	5–6
- Provides some knowledge of mutation in general terms and its potential to affect whole organisms OR - Provides some understanding of germ line mutation and reasons for its impact on whole organisms OR - Provides some understanding of transposons in their disruption of genes in genomes	3–4
- Provides some knowledge of mutations in general terms OR - Provides some relevant information on germ line mutations OR - Provides some relevant information on transposons	1–2

Question 34 (a) (i)

Criteria	Marks
Provides TWO unique skeletal features	2
Provides ONE unique skeletal feature	1

Question 34 (a) (ii)

Criteria	Marks
Provides one named non-skeletal feature and its associated correct evolution advantage	2
Provides some information about a non-skeletal feature	1

Question 34 (b) (i)

Criteria	Marks
- Comparatively describes the data from 2008 - Relates cause and effect for TWO relevant conclusions drawn from the data about m-DNA	3
- Comparatively describes the data from 2008 - Relates cause and effect for ONE relevant conclusion drawn from the data about m-DNA	2
- Describes the data from 2008 OR - Relates cause and effect for ONE relevant conclusion drawn from the data about m-DNA	1

Question 34 (b) (ii)

Criteria	Marks
- Comparatively describes the data from 2010 - Relates cause and effect for TWO relevant conclusions drawn from the data about migration and breeding	3
- Comparatively describes the data from 2010 - Relates cause and effect for ONE relevant conclusion drawn from the data about migration and breeding	2
- Describes the data from 2010 OR - Relates cause and effect for ONE relevant conclusion drawn from the data about migration and breeding	1

Question 34 (c)

Criteria	Marks
- Provides a clear distinction between polymorphism and clinal gradation as they relate to populations - Identifies the events that lead to polymorphism and clinal gradation and links to phenotype survival - Includes at least one human phenotype as an example	6–7
- Provides an understanding of the differences between polymorphism and clinal gradation - Includes at least one human phenotype as an example OR - Identifies the events that lead to polymorphism and clinal gradation - Includes at least one human phenotype as an example	4–5
- Provides an understanding of polymorphism or clinal gradation - Includes a human example OR - Identifies the events that lead to polymorphism or clinal gradation - Includes a human example	2–3
Provides some statement relevant to polymorphism or clinal gradation	1

Question 34 (d) (i)

Criteria	Marks
Names TWO hominids and links correct fossil evidence to technology and cultural development	2
- Names TWO fossil hominids and links to correct fossil evidence OR - Names ONE fossil hominid and links correct fossil evidence to technology and cultural development	1

Question 34 (d) (ii)

Criteria	Marks
- Communicates understanding of biological evolution - Identifies at least TWO relevant technologies developed by modern humans - Relates in detail the possible effect of the use of each technology on human biological evolution - Articulates ideas in a coherent and logical manner and includes the correct use of scientific principles and ideas	5–6
- Communicates understanding of biological evolution - Identifies at least ONE relevant technology developed by modern humans - Relates in detail the possible effect of the use of ONE technology on human biological evolution OR - Identifies at least TWO relevant technologies developed by modern humans - Relates the possible effect of the use of each technology on human biological evolution	3–4
- Identifies at least ONE relevant technology developed by modern humans - Relates the possible effect of use of ONE technology on human biological evolution	2
Provides some information about a relevant technology used by modern humans	1

Question 35 (a) (i)

Criteria	Marks
Identifies a product of photosynthesis and its related source	2
Identifies either a product of photosynthesis OR source of photosynthesis	1

Question 35 (a) (ii)

Criteria	Marks
Identifies TWO non-renewable resources and links their replacement with photosynthesis	2
- Identifies ONE non-renewable resource and links its replacement with photosynthesis OR - Identifies TWO non-renewable resources but does not relate to photosynthesis	1

Question 35 (b) (i)

Criteria	Marks
• Diagram provides a detailed description of the experimental set-up	2
• Diagram conveys an adequate description of the experimental set-up	1

Question 35 (b) (ii)

Criteria	Marks
• Identifies that volume of gas is measured and relates it to a variable	2
• Identifies that a gas is being measured	1

Question 35 (b) (iii)

Criteria	Marks
• Identifies how the model could be changed to vary carbon dioxide and keep the light levels constant	2
• Identifies how carbon dioxide levels could be changed	1

Question 35 (c) (i)

Criteria	Marks
• Relates the diagram to the storage and release of energy in a biological context	2
• Gives a biological function of ATP OR • Relates the diagram to the storage and release of energy	1

Question 35 (c) (ii)

Criteria	Marks
Provides extensive and detailed knowledge and superior understanding of the series of steps associated with photosynthesis that lead to the production of energy stores	5
Provides extensive knowledge and understanding of the series of steps associated with photosynthesis that lead to the production of energy stores	4
- Provides some knowledge of the series of steps associated with photosynthesis OR - Provides some knowledge about the production of ATP OR - Provides some knowledge about the production of NADPH	2–3
Provides some statement relevant to the light-dependent reaction pathway	1

Question 35 (d)

Criteria	Marks
- Provides extensive knowledge of the historical development of biochemical pathway investigation - Includes names of specific scientists and describes what they did - Provides knowledge of different radioisotopes and their half-lives and is able to relate this to use in biochemical pathway tracing. - Articulates ideas in a coherent and logical manner and includes the correct use of scientific principles and ideas	7–8
- Provides some knowledge of the historical development of pathway investigation - Includes names of specific scientists and describes what they did - Provides knowledge of a radioisotope and how it could be used in biochemical pathway tracing - Articulates ideas in a coherent and logical manner and includes the correct use of scientific principles and ideas	5–6
- Provides some knowledge of the historical development of pathway investigation - Includes the name of a specific scientist and describes what he/she did - Provides knowledge of a radioisotope	3–4
Shows some understanding of the historical development of the understanding of photosynthesis	2
Provides some statement relevant to radioisotopes	1

Biology

2012 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.3.5.2.2, 9.3.4.2.2, 9.3.5.3.3.2	H3, H7, H9
2	1	9.3.1.2.2, 9.3.1.2.3, 9.1.14.1.a, 9.1.14.1.c	H10, H14
3	1	9.4.4.2.1, 9.4.4.2.4, 9.4.5.2.1	H6
4	1	9.4.3.2.2, 9.1.11.1(c), 9.1.14.1(d)	H6 H14
5	1	9.4.6.2.2,	H6, H9
6	1	9.4.5.2.3	H6
7	1	9.4.3.2.3, 9.4.3.3.4	H4
8	1	9.3.3.2.5, 9.3.3.2.6, 9.3.3.2.9, 9.3.3.3.1, 9.3.3.3.3	H9
9	1	9.2.2.3.6	H6
10	1	9.2.3.2.4, 9.2.3.2.5, 9.1.14.1d, 9.1.14.1f	H6, H14
11	1	9.3.5.2.1, 9.3.5.2.2, 9.3.2.3.3	H9
12	1	9.2.2.2.4, 9.2.3.3.1, 9.2.3.2.6	H6
13	1	9.4.4.2.2, 9.4.4.2.3, 9.4.5.2.1, 9.4.5.2.4	H4, H6
14	1	9.2.3.2.3, 9.2.3.3.4	H6, H7
15	1	9.2.3.2.5, 9.2.3.3.2, 9.2.3.2.7	H6
16	1	9.4.2.3.1, 9.1.11.2b, 9.1.12.4d, 9.1.14.1(a)	H2, H11, H12, H14
17	1	9.3.2.2.5	H6, H9
18	1	9.2.1.2.9, 9.2.1.3.3	H7, H10
19	1	9.2.2.2.1, 9.2.2.2.2, 9.2.2.2.5	H6, H10
20	1	9.3.1.3.1, 9.1.14.1a, 9.1.14.1b, 9.1.14.1f	H10, H14
Section I Part B			
21 (a)	2	9.2.2.3.2, 9.1.11.3.a, 9.1.11.1b, 9.1.13.1e	H6
21 (b)	3	9.2.2.3.2	H6
22 (a)	2	9.1.12.3.c, 9.1.13.1.e, 9.2.2.3.1	H6, H13
22 (b)	3	9.1.13.1.f	H2, H13
22 (c)	2	9.1.13.1.f, 9.1.13.1.g	H13
23 (a)	1	9.4.6.2.2, 9.1.11.1b	H9, H11
23 (b) (i)	2	9.3.2.3.1, 9.1.13.1e	H9, H13
23 (b) (ii)	2	9.3.2.2.6, 9.1.14.1b	H9, H14
24 (a)	1	9.2.1.3.1, 9.1.11.2.a, 9.1.11.2.b	H6, H11
24 (b)	2	9.2.1.3.1, 9.1.11.1b	H6, H11

Question	Marks	Content	Syllabus outcomes
24 (c)	2	9.2.1.2.3, 9.2.1.2.4, 9.2.1.2.7, 9.2.1.2.8	H6
25 (a)	4	9.3.4.2.5, 9.3.4.3.2, 9.3.4.3.3, 9.1.14.1g	H6, H9, H14
25 (b)	2	9.1.12.4d, 9.1.12.4e, 9.1.12.4f	H6, H9, H12
26	5	9.4.6.3.1, 9.4.6.2.1, 9.1.11.1b, 9.1.11.1a	H2, H11
27	4	9.4.7.1, 9.4.7.2.1, 9.4.7.3.2, 9.4.7.3.3, 9.1.14.2b, 9.1.14.2c	H4
28	5	9.3.4.1.1, 9.3.4.2.1, 9.3.4.2.4, 9.3.3.2.5	H9, H10
29 (a)	3	9.3.3.2.5, 9.3.3.3.1, 9.1.13.1.e, 9.1.14.1.1, 9.1.14.1a, 9.1.14.3b	H9, H13, H14
29 (b)	2	9.3.3.2.5, 9.3.3.3.1	H9
30	8	9.4.3.1, 9.4.5.1, 9.4.6.1, 9.4.7.1	H1, H3, H4:
Section II			
Question 31 — Communication			
(a) (i)	2	9.5.6.2.2, 9.1.13.1e	H6, H13
(a) (ii)	2	9.5.6.2.2, 9.5.6.3.1, 9.5.6.2.4	H6
(b) (i)	2	9.5.2.2.1, 9.5.3.2.3, 9.5.3.3.1, 9.5.3.3.2, 9.1.11.3a, 9.1.14.1f	H6, H11, H14
(b) (ii)	2	9.5.2.2.1, 9.5.3.2.3, 9.5.3.2.4, 9.5.3.3.1, 9.5.3.3.2, 9.1.11.1b	H6, H11
(b) (iii)	2	9.5.2.2.1, 9.5.3.2.3, 9.5.3.2.4, 9.5.3.3.1, 9.5.3.3.2, 9.1.14.3c, 9.1.14.1f	H6, H14
(c) (i)	2	9.5.1.1, 9.5.2.2.1, 9.5.4.3.2	H1, H4, H6, H14
(c) (ii)	5	9.5.6.3.2, 9.5.6.2.1, 9.1.14.3b, 9.5.7.1	H1, H4, H6, H14
(d)	8	9.5.3.2.5, 9.5.3.3.3, 9.5.6.3.3, 9.5.7.2.5, 9.5.7.3.2, 9.1.14.1g, 9.1.14.3b, 9.1.14.3d	H2, H4, H14
Section II			
Question 32 — Biotechnology			
(a)	4	9.6.2.2.1, 9.1.13.1.b	H6, H13
(b) (i)	3	9.6.5.3.3	H9
(b) (ii)	3	9.6.5.3.3, 9.1.13.1.e, 9.6.5.3.2	H9, H13
(c) (i)	2	9.6, 9.6.6.3.1	H2, H4, H6
(c) (ii)	5	9.6.1, 9.6.6.2.4, 9.6.6.3.1	H2, H4, H6
(d)	8	9.1.14.3, 9.6.1, 9.6.2, 9.6.3, 9.6.4, 9.6.5, 9.6.6	H1, H2, H3, H4, H14

Section II			
Question 33 — Genetics: The Code Broken?			
(a)	4	9.7.7.2.2, 9.7.7.2.3, 9.7.7.3.2, 9.1.12.3c	H 9, H12
(b) (i)	3	9.7.1.3.1 9.1.14.1f	H2, H14
(b) (ii)	3	9.7.1.2.1, 9.7.1.3.2	H9
(c) (i)	2	9.7.4.1, 9.7.4.2.1, 9.7.4.2.2, 9.7.4.3.1	H1, H3
(c) (ii)	5	9.7.3.2.4, 9.7.4.3.1, 9.7.4.2.4	H1, H3
(d)	8	9.7.6.2.3, 9.7.6.2.1, 9.7.7.3.1	H4
Section II			
Question 34 — The Human Story			
(a) (i)	2	9.8.1.2.5, 9.8.1.2.6	H10
(a) (ii)	2	9.8.5.3.1	H10
(b) (i)	3	9.8.2.3.3 9.1.14.1.b, 9.1.14.1.c, 9.1.14.1.d	H6, H10, H14
(b) (ii)	3	9.8.3.2.2, 9.8.3.3.3, 9.8.5.3.1, 9.1.14.1.b, 9.1.14.1.c, 9.1.14.1.d	H6, H10, H14
(c)	7	9.8.4.2.2, 9.8.4.3.1, 9.8.4.2.1	H2, H10
(d) (i)	2	9.8.5, 9.8.3.2.1, 9.8.3.2.2, 9.8.3.3.1	H4, H10
(d) (ii)	6	9.8.6, 9.8.6.2.1, 9.8.6.3.2	H4, H10
Section II			
Question 35 — Biochemistry			
(a) (i)	2	9.9.1.3.1	H6
(a) (ii)	2	9.9.1.2.1, 9.9.1.3.2	H5
(b) (i)	2	9.9.3.3.1, 9.1.11.3a 9.1.13.1.e	H2, H6, H11, H13
(b) (ii)	2	9.9.3.3.1, 9.1.11.1b	H6, H11
(b) (iii)	2	9.9.3.3.1, 9.1.11.1a, 9.1.12.1.a, 9.1.12.2a, 9.1.14.3a	H2, H6, H11, H12, H14
(c) (i)	2	9.9.7.2.1, 9.9.7.2.2, 9.9.7.3.1, 9.1.13.1e	H6, H13
(c) (ii)	5	9.9.4.2.3, 9.9.4.2.4, 9.9.4.2.5, 9.9.4.2.6, 9.9.4.3.2	H6
(d)	8	9.9.5	H1, H2, H6